Fiscal Year:	FY 2018	**Task Last Updated:**	FY 02/27/2020
PI Name:	Globus, Ruth Ph.D.		
Project Title:	Skeletal Responses to Long-Duration Simulated Microgravity in Male and Female Rats		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Ruth.K.Globus@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-5247
Organization Name:	NASA Ames Research Center		
PI Address 1:	Bone and Signaling Laboratory		
PI Address 2:	Space Biosciences Research Branch		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	08/01/2015	**End Date:**	09/30/2018
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	Biospecimen Sharing NOTE: Extended to 9/30/2018 (original end date was 6/30/2017) per PI/ARC (Ed., 5/8/18) NOTE: Extended to 9/30/2017 (original end date was 6/30/2017) per A. Ronca/ARC (Ed., 5/4/17) NOTE: End date should be 6/30/2017, per A. Ronca/NASA ARC (Ed., 5/11/16) NOTE: Extended to 12/1/2016, per A. Ronca/NASA ARC (Ed., 5/10/16) NOTE: change in period of performance to 8/1/2015-7/31/2016, per A. Ronca/NASA Ames (original period of performance was 6/25/14-9/30/16)--Ed., 3/27/15		
Key Personnel Changes/Previous PI:	Changes implemented, reported in February 2020 report: Candice Tahimic - Changed from collaborator to Co-Investigator ; Gabriella Loots - Added as Co-Investigator.		

COI Name (Institution):	Blaber, Elizabeth Ph.D. (NASA Ames Research Center) Shirazi-Fard, Yasaman Ph.D. (NASA Ames Research Center) Tahimic, Candice GT Ph.D. (NASA Ames Research Center / KBR) Loots, Gabriela Ph.D. (Lawrence Livermore National Laboratory) Hogan, Harry Ph.D. (Texas A&M University)
Grant/Contract No.:	Internal Project
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Prolonged bedrest, musculoskeletal disuse, and spaceflight cause complex metabolic adaptations and cardiovascular changes. In addition to exposure to microgravity, the International Space Station (ISS) poses a unique and complex set of physical challenges to physiologic homeostasis; these include confinement, altered gas composition (e.g., elevated ambient CO2), and space radiation. In rodents, short duration spaceflight causes pronounced changes in the expression of proteins and genes related to oxidative metabolism, implicating compensatory responses. The full extent and mechanisms of cardiovascular changes over the long term as a function of age or gender are not understood. We hypothesize that the unique environment of space alters cellular redox balance leading to oxidative damage and ultimately, degenerative changes in cardiac tissue of both males and female rats. This proposed project will test three Specific Aims, taking advantage of the unique tissue sharing opportunity afforded by the University of California Davis experiment using rats which includes hindlimb unloading, reloading, of both males and females, young and older, using biochemical and gene expression methods. This research will yield new understanding about how the cardiovascular system responds to conditions that simulate weightlessness.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from this research will identify basic molecular mechanisms that may underlie degenerative skeletal changes that arise due to prolonged musculoskeletal disuse on Earth. The findings are relevant to osteoporosis, a common and often devastating skeletal diseases. On Earth, osteoporosis can result from aging, hormonal changes, immobilization, prolonged inactivity, and bedrest. In addition, osteoporosis affects the ability of skeletal tissue to recover from fractures. Identification of specific mechanisms that contribute to the initiation and progression of osteoporosis with the hindlimb unloading model of spaceflight will contribute to the development of new countermeasures.
Task Progress:	Spaceflight can profoundly affect skeletal structure and the expression levels of genes related to redox signaling, although the molecular links between oxidative stress, bone resorption, and mechanical unloading of skeletal tissue are not yet fully understood. Nor do we understand how age, sex, and time influence the skeletal responses to weightlessness. We hypothesized that gene expression changes in oxidative defense-related pathways and pro-osteoclastogenic signals are invoked for skeletal adaptation to long duration simulated weightlessness by hindlimb unloading (HU) and recovery. Results obtained have provided new insight into how simulated weightlessness by hindlimb unloading influences the skeleton while addressing our hypotheses. HU caused structural deficits sooner in young rats (3 mo old) vs adult rats (9 mo old). In young rats, complete recovery occurred in structural and mechanical properties of the cortical mid-shaft after from 90 d of HU followed by 90 d of ambulation. Analysis of the femoral shaft from young male rats unloaded for 90 d showed regulation of pathways related to bone resorption and formation, but not to those of cellular redox or oxidative stress at this late timepoint. In sum, simulated weightlessness by hindlimb unloading led to structural and mechanical deficits that displayed specific dependencies on age, sex, HU duration, and skeletal site. Findings on the effects of simulated weightlessness, with particular relevance to Human Research Program (HRP) and human space travel include: • Gene expression: Long duration (90 d of HU) profiles show networks enriched for genes related to resorption, formation, and other signaling processes, but unrelated to oxidative stress. These findings implicate a state of elevated bone turnover even at this very late stage of simulated weightlessness. • Site specificity: measurement of cancellous bone structure at sites more distal to each other within the same bone show weaker correlations than do measurements at more proximal sites. • Sex dependency: young males appeared more sensitive to skeletal effects of HU than young females at an early HU stage (14 d) with respect to decrements in cancellous and cortical structure and mechanical properties. • Age dependency: at an early HU stage, adult males showed a decrement in mechanical properties (femoral shaft in 3-point bending) whereas growing males did not until 90 d of HU. Thus, adult males may be more sensitive than young to HU. • Recovery capacity: Cortical bone recovered with prolonged re-ambulation (after a similar period of HU), whereas cancellous tissue remained low, resembling the findings in astronauts. • Because of possible contributions of modeling during growth of young animals to observed responses to simulated weightlessness, our findings emphasize the importance of studying the adult rather than the young rodent.
Bibliography Type:	Description: (Last Updated: 06/04/2025)
Abstracts for Journals and Proceedings	Elizondo J, Tahimic CGT, O'Neil N, Torres S, Hogan H, Globus RK, Shirazi-Fard Y. "Time-dependent effects of long duration simulated weightlessness in the rat skeleton." Poster presentation at the 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020

Abstracts for Journals and Proceedings	Tahimic CGT, Torres S, O'Neil N, Schreurs AS, Blaber E, Alwood JS, Hogan H, Globus RK, Shirazi-Fard Y. "Time-dependent effects of long duration simulated weightlessness in the rat skeleton." Oral presentation at the 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. Abstracts. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019
Abstracts for Journals and Proceedings	Tahimic CGT, Torres S, Adams J, Schreurs AS, Alwood JS, Shirazi-Fard Y, Globus RK. "Time-dependent effects of long duration simulated weightlessness in the rat skeleton." Poster presented at the 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. Abstracts. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
Abstracts for Journals and Proceedings	Adams J, Torres S, Schreurs AS, Alwood JS, Shirazi-Fard Y, Tahimic CGT, Globus RK. "Skeletal Responses to Long-Duration Simulated Microgravity in Rats." Poster presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. Abstracts. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Articles in Peer-reviewed Journals	Tahimic CGT, Steczina S, Sebastian A, Hum NR, Abegaz M, Terada M, Cimini M, Goukassian DA, Schreurs A-S, Hoban-Higgins TM, Fuller CA, Loots GG, Globus RK, Shirazi-Fard Y. "Simulated microgravity alters gene regulation linked to immunity and cardiovascular disease." Genes. 2024 Jul 24;15(8):975. https://doi.org/10.3390/genes15080975 ; PMID: 39202335; PMCID: PMC11353732 , Jul-2024